# CLASS 9 — MATHS · FORMULA SHEET

Chapter 14: **Surface Area and Volume** | Quick revision, one page

## Cuboid and cube

- Cuboid $(l, b, h)$: volume $= lbh$; TSA $= 2(lb + bh + hl)$; LSA $= 2h(l + b)$; diagonal $= \sqrt{l^2 + b^2 + h^2}$.
- Cube (side $a$): volume $= a^3$; TSA $= 6a^2$; LSA $= 4a^2$; diagonal $= a\sqrt{3}$.

## Cylinder

- Volume $= \pi r^2 h$; CSA $= 2\pi r h$; TSA $= 2\pi r(r + h)$.

## Cone

- Slant height $l = \sqrt{r^2 + h^2}$.
- Volume $= \frac{1}{3}\pi r^2 h$; CSA $= \pi r l$; TSA $= \pi r(l + r)$.

## Sphere and hemisphere

- Sphere: volume $= \frac{4}{3}\pi r^3$; surface area $= 4\pi r^2$.
- Hemisphere: volume $= \frac{2}{3}\pi r^3$; CSA $= 2\pi r^2$; TSA $= 3\pi r^2$.

## Pyramid and recasting

- Pyramid: volume $= \frac{1}{3} \times$ (base area) $\times$ height; lateral area $=$ sum of the areas of the triangular faces.
- Recasting (melting one solid into another): volume is unchanged; number of small solids $= \dfrac{\text{volume of large solid}}{\text{volume of one small solid}}$.
- Units: area in cm$^2$ or m$^2$, volume in cm$^3$ or m$^3$; $1\ \text{m}^3 = 1000$ litres, $1000\ \text{cm}^3 = 1$ litre.

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